

Having shown the existence of $K(G, n)$'s, we now consider the uniqueness question, which has the nicest possible answer:

Proposition 4.30. *The homotopy type of a CW complex $K(G, n)$ is uniquely determined by G and n .*

The proof will be based on a more technical statement:

Lemma 4.31. *Let X be a CW complex of the form $(\bigvee_{\alpha} S_{\alpha}^n) \cup_{\beta} e_{\beta}^{n+1}$ for some $n \geq 1$. Then for every homomorphism $\psi: \pi_n(X) \rightarrow \pi_n(Y)$ with Y path-connected there exists a map $f: X \rightarrow Y$ with $f_* = \psi$.*

Proof: To begin, let f send the natural basepoint of $\bigvee_{\alpha} S_{\alpha}^n$ to a chosen basepoint $y_0 \in Y$. Extend f over each sphere S_{α}^n via a map representing $\psi([i_{\alpha}])$ where i_{α} is the inclusion $S_{\alpha}^n \hookrightarrow X$. Thus for the map $f: X^n \rightarrow Y$ constructed so far we have $f_*([i_{\alpha}]) = \psi([i_{\alpha}])$ for all α , hence $f_*([\varphi]) = \psi([\varphi])$ for all basepoint-preserving maps $\varphi: S^n \rightarrow X^n$ since the i_{α} 's generate $\pi_n(X^n)$. To extend f over a cell e_{β}^{n+1} all we need is that the composition of the attaching map $\varphi_{\beta}: S^n \rightarrow X^n$ for this cell with f be nullhomotopic in Y . But this composition $f\varphi_{\beta}$ represents $f_*([\varphi_{\beta}]) = \psi([\varphi_{\beta}])$, and $\psi([\varphi_{\beta}]) = 0$ because $[\varphi_{\beta}]$ is zero in $\pi_n(X)$ since φ_{β} is nullhomotopic in X via the characteristic map of e_{β}^{n+1} . Thus we obtain an extension $f: X \rightarrow Y$. This has $f_* = \psi$ since the elements $[i_{\alpha}]$ generate $\pi_n(X^n)$ and hence also $\pi_n(X)$ by cellular approximation. $\square$

Proof of 4.30: Suppose K and K' are $K(G, n)$ CW complexes. Since homotopy equivalence is an equivalence relation, there is no loss of generality if we assume K is a particular $K(G, n)$, namely one constructed from a space X as in the lemma by attaching cells of dimension $n+2$ and greater. By the lemma there is a map $f: X \rightarrow K'$ inducing an isomorphism on π_n . To extend this f over K we proceed inductively. For each cell e^{n+2} , the composition of its attaching map with f is nullhomotopic in K' since $\pi_{n+1}(K') = 0$, so f extends over this cell. The same argument applies for all the higher-dimensional cells in turn. The resulting $f: K \rightarrow K'$ is a homotopy equivalence since it induces isomorphisms on all homotopy groups. $\square$

The Hurewicz Theorem

Using the calculations of homotopy groups done above we can easily prove the simplest and most often used cases of the Hurewicz theorem:

Theorem 4.32. *If a space X is $(n-1)$ -connected, $n \geq 2$, then $\tilde{H}_i(X) = 0$ for $i < n$ and $\pi_n(X) \approx H_n(X)$. If a pair (X, A) is $(n-1)$ -connected, $n \geq 2$, with A simply-connected and nonempty, then $H_i(X, A) = 0$ for $i < n$ and $\pi_n(X, A) \approx H_n(X, A)$.*

Thus the first nonzero homotopy and homology groups of a simply-connected space occur in the same dimension and are isomorphic. One cannot expect any nice

relationship between $\pi_i(X)$ and $H_i(X)$ beyond this. For example, S^n has trivial homology groups above dimension n but many nontrivial homotopy groups in this range when $n \geq 2$. In the other direction, Eilenberg–MacLane spaces such as $\mathbb{C}P^\infty$ have trivial higher homotopy groups but many nontrivial homology groups.

The theorem can sometimes be used to compute $\pi_2(X)$ if X is a path-connected space that is nice enough to have a universal cover. For if $\tilde{X}$ is the universal cover, then $\pi_2(X) \approx \pi_2(\tilde{X})$ and the latter group is isomorphic to $H_2(\tilde{X})$ by the Hurewicz theorem. So if one can understand $\tilde{X}$ well enough to compute $H_2(\tilde{X})$, one can compute $\pi_2(X)$.

In the part of the theorem dealing with relative groups, notice that X must be simply-connected as well as A since (X, A) is 1-connected by hypothesis. There is a more general version of the relative Hurewicz theorem given later in Theorem 4.37 that allows A and X to be nonsimply-connected, but this requires $\pi_n(X, A)$ to be replaced by a certain quotient group.

Proof: We may assume X is a CW complex and (X, A) is a CW pair by taking CW approximations to X and (X, A) . For CW pairs the relative case then reduces to the absolute case since $\pi_i(X, A) \approx \pi_i(X/A)$ for $i \leq n$ by Proposition 4.28, while $H_i(X, A) \approx \tilde{H}_i(X/A)$ for all i by Proposition 2.22.

In the absolute case we can apply Corollary 4.16 to replace X by a homotopy equivalent CW complex with $(n-1)$ -skeleton a point, hence $\tilde{H}_i(X) = 0$ for $i < n$. To show $\pi_n(X) \approx H_n(X)$, we can further simplify by throwing away cells of dimension greater than $n+1$ since these have no effect on π_n or H_n . Thus X has the form $(\bigvee_\alpha S_\alpha^n) \cup_\beta e_\beta^{n+1}$. We may assume the attaching maps φ_β of the cells e_β^{n+1} are basepoint-preserving since this is what the proof of Corollary 4.16 gives. Example 4.29 then applies to compute $\pi_n(X)$ as the cokernel of the boundary map $\pi_{n+1}(X, X^n) \rightarrow \pi_n(X^n)$, a map $\bigoplus_\beta \mathbb{Z} \rightarrow \bigoplus_\alpha \mathbb{Z}$. This is the same as the cellular boundary map $d: H_{n+1}(X^{n+1}, X^n) \rightarrow H_n(X^n, X^{n-1})$ since for a cell e_β^{n+1} , the coefficients of de_β^{n+1} are the degrees of the compositions $q_\alpha \varphi_\beta$ where q_α collapses all n -cells except e_α^n to a point, and the isomorphism $\pi_n(S^n) \approx \mathbb{Z}$ in Corollary 4.25 is given by degree. Since there are no $(n-1)$ -cells, we have $H_n(X) \approx \text{Coker } d$. $\square$

Since homology groups are usually more computable than homotopy groups, the following version of Whitehead's theorem is often easier to apply:

Corollary 4.33. *A map $f: X \rightarrow Y$ between simply-connected CW complexes is a homotopy equivalence if $f_*: H_n(X) \rightarrow H_n(Y)$ is an isomorphism for each n .*

Proof: After replacing Y by the mapping cylinder M_f we may take f to be an inclusion $X \hookrightarrow Y$. Since X and Y are simply-connected, we have $\pi_1(Y, X) = 0$. The relative Hurewicz theorem then says that the first nonzero $\pi_n(Y, X)$ is isomorphic to the first nonzero $H_n(Y, X)$. All the groups $H_n(Y, X)$ are zero from the long exact sequence of homology, so all the groups $\pi_n(Y, X)$ also vanish. This means that the inclusion

$X \hookrightarrow Y$ induces isomorphisms on all homotopy groups, and therefore this inclusion is a homotopy equivalence. $\square$

Example 4.34: Uniqueness of Moore Spaces. Let us show that the homotopy type of a CW complex Moore space $M(G, n)$ is uniquely determined by G and n if $n > 1$, so $M(G, n)$ is simply-connected. Let X be an $M(G, n)$ as constructed in Example 2.40 by attaching $(n+1)$ -cells to a wedge sum of n -spheres, and let Y be any other $M(G, n)$ CW complex. By Lemma 4.31 there is a map $f: X \rightarrow Y$ inducing an isomorphism on π_n . If we can show that f also induces an isomorphism on H_n , then the preceding corollary will imply the result.

One way to show that f induces an isomorphism on H_n would be to use a more refined version of the Hurewicz theorem giving an isomorphism between π_n and H_n that is natural with respect to maps between spaces, as in Theorem 4.37 below. However, here is a direct argument which avoids naturality questions. For the mapping cylinder M_f we know that $\pi_i(M_f, X) = 0$ for $i \leq n$. If this held also for $i = n+1$ then the relative Hurewicz theorem would say that $H_i(M_f, X) = 0$ for $i \leq n+1$ and hence that f_* would be an isomorphism on H_n . To make this argument work, let us temporarily enlarge Y by attaching $(n+2)$ -cells to make π_{n+1} zero. The new mapping cylinder M_f then has $\pi_{n+1}(M_f, X) = 0$ from the long exact sequence of the pair. So for the enlarged Y the map f induces an isomorphism on H_n . But attaching $(n+2)$ -cells has no effect on H_n , so the original $f: X \rightarrow Y$ had to be an isomorphism on H_n .

It is certainly possible for a map of nonsimply-connected spaces to induce isomorphisms on all homology groups but not on homotopy groups. Nonsimply-connected acyclic spaces, for which the inclusion of a point induces an isomorphism on homology, exhibit this phenomenon in its purest form. Perhaps the simplest nontrivial acyclic space is the 2-dimensional complex constructed in Example 2.38 with fundamental group $\langle a, b \mid a^5 = b^3 = (ab)^2 \rangle$ of order 120.

It is also possible for a map between spaces with abelian fundamental groups to induce isomorphisms on homology but not on higher homotopy groups, as the next example shows.

Example 4.35. We construct a space $X = (S^1 \vee S^n) \cup e^{n+1}$, for arbitrary $n > 1$, such that the inclusion $S^1 \hookrightarrow X$ induces an isomorphism on all homology groups and on π_i for $i < n$, but not on π_n . From Example 4.27 we have $\pi_n(S^1 \vee S^n) \approx \mathbb{Z}[t, t^{-1}]$. Let X be obtained from $S^1 \vee S^n$ by attaching a cell e^{n+1} via a map $S^n \rightarrow S^1 \vee S^n$ corresponding to $2t - 1 \in \mathbb{Z}[t, t^{-1}]$. By looking in the universal cover we see that $\pi_n(X) \approx \mathbb{Z}[t, t^{-1}]/(2t - 1)$, where $(2t - 1)$ denotes the ideal in $\mathbb{Z}[t, t^{-1}]$ generated by $2t - 1$. Note that setting $t = 1/2$ embeds $\mathbb{Z}[t, t^{-1}]/(2t - 1)$ in $\mathbb{Q}$ as the subring $\mathbb{Z}[1/2]$ consisting of rationals with denominator a power of 2. From the long exact sequence of homotopy groups for the $(n-1)$ -connected pair (X, S^1) we see that the inclusion

$S^1 \hookrightarrow X$ induces an isomorphism on π_i for $i < n$. The fact that this inclusion also induces isomorphisms on all homology groups can be deduced from cellular homology. The key point is that the cellular boundary map $H_{n+1}(X^{n+1}, X^n) \rightarrow H_n(X^n, X^{n-1})$ is an isomorphism since the degree of the composition of the attaching map $S^n \rightarrow S^1 \vee S^n$ of e^{n+1} with the collapse $S^1 \vee S^n \rightarrow S^n$ is $2 - 1 = 1$.

This example relies heavily on the nontriviality of the action of $\pi_1(X)$ on $\pi_n(X)$, so one might ask whether the simple-connectivity assumption in Corollary 4.33 can be weakened to trivial action of π_1 on all π_n 's. This is indeed the case, as we will show in Proposition 4.74.

The form of the Hurewicz theorem given above asserts merely the existence of an isomorphism between homotopy and homology groups, but one might want a more precise statement which says that a particular map is an isomorphism. In fact, there are always natural maps from homotopy groups to homology groups, defined in the following way. Thinking of $\pi_n(X, A, x_0)$ for $n > 0$ as homotopy classes of maps $f: (D^n, \partial D^n, s_0) \rightarrow (X, A, x_0)$, the **Hurewicz map** $h: \pi_n(X, A, x_0) \rightarrow H_n(X, A)$ is defined by $h([f]) = f_*(\alpha)$ where α is a fixed generator of $H_n(D^n, \partial D^n) \approx \mathbb{Z}$ and $f_*: H_n(D^n, \partial D^n) \rightarrow H_n(X, A)$ is induced by f . If we have a homotopy $f \simeq g$ through maps $(D^n, \partial D^n, s_0) \rightarrow (X, A, x_0)$, or even through maps $(D^n, \partial D^n) \rightarrow (X, A)$ not preserving the basepoint, then $f_* = g_*$, so h is well-defined.

Proposition 4.36. *The Hurewicz map $h: \pi_n(X, A, x_0) \rightarrow H_n(X, A)$ is a homomorphism, assuming $n > 1$ so that $\pi_n(X, A, x_0)$ is a group.*

Proof: It suffices to show that for maps $f, g: (D^n, \partial D^n) \rightarrow (X, A)$, the induced maps on homology satisfy $(f + g)_* = f_* + g_*$, for if this is the case then $h([f + g]) = (f + g)_*(\alpha) = f_*(\alpha) + g_*(\alpha) = h([f]) + h([g])$. Our proof that $(f + g)_* = f_* + g_*$ will in fact work for any homology theory.

Let $c: D^n \rightarrow D^n \vee D^n$ be the map collapsing the equatorial D^{n-1} to a point, and let $q_1, q_2: D^n \vee D^n \rightarrow D^n$ be the quotient maps onto the two summands, collapsing the other summand to a point. We then have a diagram

$$\begin{array}{ccccc} H_n(D^n, \partial D^n) & \xrightarrow{c_*} & H_n(D^n \vee D^n, \partial D^n \vee \partial D^n) & \xrightarrow{(f \vee g)_*} & H_n(X, A) \\ & & q_{1*} \oplus q_{2*} \downarrow \approx & & \\ & & H_n(D^n, \partial D^n) \oplus H_n(D^n, \partial D^n) & & \end{array}$$

The map $q_{1*} \oplus q_{2*}$ is an isomorphism with inverse $i_{1*} + i_{2*}$ where i_1 and i_2 are the inclusions of the two summands $D^n \hookrightarrow D^n \vee D^n$. Since $q_1 c$ and $q_2 c$ are homotopic to the identity through maps $(D^n, \partial D^n) \rightarrow (D^n, \partial D^n)$, the composition $(q_{1*} \oplus q_{2*})c_*$ is the diagonal map $x \mapsto (x, x)$. From the equalities $(f \vee g)i_1 = f$ and $(f \vee g)i_2 = g$ we deduce that $(f \vee g)_*(i_{1*} + i_{2*})$ sends $(x, 0)$ to $f_*(x)$ and $(0, x)$ to $g_*(x)$, hence (x, x) to $f_*(x) + g_*(x)$. Thus the composition across the top of the diagram is

$x \mapsto f_*(x) + g_*(x)$. On the other hand, $f + g = (f \vee g)c$, so this composition is also $(f + g)_*$. $\square$

There is also an absolute Hurewicz map $h: \pi_n(X, x_0) \rightarrow H_n(X)$ defined in a similar way by setting $h([f]) = f_*(\alpha)$ for $f: (S^n, s_0) \rightarrow (X, x_0)$ and α a chosen generator of $H_n(S^n)$. For example, if $X = S^n$ then $f_*(\alpha)$ is $(\deg f)\alpha$ by the definition of degree, so we can view h in this case as the degree map $\pi_n(S^n) \rightarrow \mathbb{Z}$, which we know is an isomorphism by Corollary 4.25. The proof of the preceding proposition is readily modified to show that the absolute h is a homomorphism for $n \geq 1$.

The absolute and relative Hurewicz maps can be combined in a diagram of long exact sequences

$$\begin{array}{ccccccc} \cdots & \longrightarrow & \pi_n(A, x_0) & \longrightarrow & \pi_n(X, x_0) & \longrightarrow & \pi_n(X, A, x_0) \longrightarrow \pi_{n-1}(A, x_0) \longrightarrow \cdots \\ & & \downarrow h & & \downarrow h & & \downarrow h \\ \cdots & \longrightarrow & H_n(A) & \longrightarrow & H_n(X) & \longrightarrow & H_n(X, A) \longrightarrow H_{n-1}(A) \longrightarrow \cdots \end{array}$$

An easy definition check which we leave to the reader shows that this diagram commutes up to sign at least. With more care in the choice of the generators α it can be made to commute exactly.

Another elementary property of Hurewicz maps is that they are natural: A map $f: (X, x_0) \rightarrow (Y, y_0)$ induces a commutative diagram as at the right, and similarly in the relative case.

$$\begin{array}{ccc} \pi_n(X, x_0) & \xrightarrow{f_*} & \pi_n(Y, y_0) \\ \downarrow h & & \downarrow h \\ H_n(X) & \xrightarrow{f_*} & H_n(Y) \end{array}$$

It is easy to construct nontrivial elements of the kernel of the Hurewicz homomorphism $h: \pi_n(X, x_0) \rightarrow H_n(X)$ if $\pi_1(X, x_0)$ acts nontrivially on $\pi_n(X, x_0)$, namely elements of the form $[y][f] - [f]$. This is because yf and f , viewed as maps $S^n \rightarrow X$, are homotopic if we do not require the basepoint to be fixed during the homotopy, so $(yf)_*(\alpha) = f_*(\alpha)$ for α a generator of $H_n(S^n)$.

Similarly in the relative case the kernel of $h: \pi_n(X, A, x_0) \rightarrow H_n(X, A)$ contains the elements of the form $[y][f] - [f]$ for $[y] \in \pi_1(A, x_0)$. For example the Hurewicz map $\pi_n(S^1 \vee S^n, S^1) \rightarrow H_n(S^1 \vee S^n, S^1)$ is the homomorphism $\mathbb{Z}[t, t^{-1}] \rightarrow \mathbb{Z}$ sending all powers of t to 1. Since the pair $(S^1 \vee S^n, S^1)$ is $(n-1)$ -connected, this example shows that the hypothesis $\pi_1(A, x_0) = 0$ in the relative form of the Hurewicz theorem proved earlier cannot be dropped.

If we define $\pi'_n(X, A, x_0)$ to be the quotient group of $\pi_n(X, A, x_0)$ obtained by factoring out the subgroup generated by all elements of the form $[y][f] - [f]$, or the normal subgroup generated by such elements in the case $n = 2$ when $\pi_2(X, A, x_0)$ may not be abelian, then h induces a homomorphism $h': \pi'_n(X, A, x_0) \rightarrow H_n(X, A)$. The general form of the Hurewicz theorem deals with this homomorphism:

Theorem 4.37. *If (X, A) is an $(n - 1)$ -connected pair of path-connected spaces with $n \geq 2$ and $A \neq \emptyset$, then $h' : \pi'_n(X, A, x_0) \rightarrow H_n(X, A)$ is an isomorphism and $H_i(X, A) = 0$ for $i < n$.*

Note that this statement includes the absolute form of the theorem by taking A to be the basepoint.

Before starting the proof of this general Hurewicz theorem we have a preliminary step:

Lemma 4.38. *If X is a connected CW complex to which cells e_α^n are attached for a fixed $n \geq 2$, forming a CW complex $W = X \cup_\alpha e_\alpha^n$, then $\pi_n(W, X)$ is a free $\pi_1(X)$ -module with basis the homotopy classes of the characteristic maps Φ_α of the cells e_α^n , provided that the map $\pi_1(X) \rightarrow \pi_1(W)$ induced by inclusion is an isomorphism. In particular, this is always the case if $n \geq 3$. In the general $n = 2$ case, $\pi_2(W, X)$ is generated by the classes of the characteristic maps of the cells e_α^2 together with their images under the action of $\pi_1(X)$.*

If the characteristic maps $\Phi_\alpha : (D^n, \partial D^n) \rightarrow (W, X)$ do not take a basepoint s_0 in ∂D^n to the basepoint x_0 in X , then they will define elements of $\pi_n(W, X, x_0)$ only after we choose change-of-basepoint paths from the points $\Phi_\alpha(s_0)$ to x_0 . Different choices of such paths yield elements of $\pi_n(W, X, x_0)$ related by the action of $\pi_1(X, x_0)$, so the basis for $\pi_n(W, X, x_0)$ is well-defined up to multiplication by invertible elements of $\mathbb{Z}[\pi_1(X)]$.

The situation when $n = 2$ and the map $\pi_1(X) \rightarrow \pi_1(W)$ is not an isomorphism is more complicated because the relative π_2 can be nonabelian in this case. Whitehead analyzed what happens here and showed that $\pi_2(W, X)$ has the structure of a ‘free crossed $\pi_1(X)$ -module.’ See [Whitehead 1949] or [Sieradski 1993].

Proof: Since $W/X = \bigvee_\alpha S_\alpha^n$, we have $\pi_n(W, X) \approx \pi_n(\bigvee_\alpha S_\alpha^n)$ when X is simply-connected, by Proposition 4.28. The conclusion of the lemma in this case is then immediate from Example 4.26.

When X is not simply-connected but the inclusion $X \hookrightarrow W$ induces an isomorphism on π_1 , then the universal cover of W is obtained from the universal cover of X by attaching n -cells lifting the cells e_α^n . If we choose one such lift $\tilde{e}_\alpha^n$ of e_α^n , then all the other lifts are the images $\gamma \tilde{e}_\alpha^n$ of $\tilde{e}_\alpha^n$ under the deck transformations $\gamma \in \pi_1(X)$. The special case proved in the preceding paragraph shows that the relative π_n for the universal cover is the free abelian group with basis corresponding to the cells $\gamma \tilde{e}_\alpha^n$. By the relative version of Proposition 4.1, the projection of the universal cover of W onto W induces an isomorphism on relative π_n ’s, so $\pi_n(W, X)$ is free abelian with basis the classes $[\gamma e_\alpha^n]$ as γ ranges over $\pi_1(X)$, or in other words the free $\pi_1(X)$ -module with basis the cells e_α^n .

It remains to consider the $n = 2$ case in general. Since both of the pairs (W, X^1) and $(X^1 \cup_\alpha e_\alpha^2, X^1)$ are 1-connected, the homotopy excision theorem implies that the

map $\pi_2(X^1 \cup_\alpha e_\alpha^2, X^1) \rightarrow \pi_2(W, X)$ is surjective. This gives a reduction to the case that X is 1-dimensional. We may also assume the 2-cells e_α^2 are attached along loops passing through the basepoint 0-cell x_0 , since this can be achieved by homotopy of the attaching maps, which does not affect the homotopy type of the pair (W, X) .

In the closure of each 2-cell e_α^2 choose an embedded disk D_α^2 which contains x_0 but is otherwise contained entirely in the interior of e_α^2 . Let $Y = X \cup_\alpha D_\alpha^2$, the wedge sum of X with the disks D_α^2 , and let $Z = W - \bigcup_\alpha \text{int}(D_\alpha^2)$, so Y and Z are 2-dimensional CW complexes with a common 1-skeleton $Y^1 = Z^1 = Y \cap Z = X \vee_\alpha \partial D_\alpha^2$. The inclusion $(W, X) \hookrightarrow (W, Z)$ is a homotopy equivalence of pairs. Homotopy excision gives a surjection $\pi_2(Y, Y^1) \rightarrow \pi_2(W, Z)$. The universal cover $\tilde{Y}$ of Y is obtained from the universal cover $\tilde{X}$ of X by taking the wedge sum with lifts $\tilde{D}_{\alpha\beta}^2$ of the disks D_α^2 . Hence we have isomorphisms

$$\begin{aligned} \pi_2(Y, Y^1) &\approx \pi_2(\tilde{Y}, \tilde{Y}^1) \quad \text{where } \tilde{Y}^1 \text{ is the 1-skeleton of } \tilde{Y} \\ &\approx \pi_2(\bigvee_{\alpha\beta} \tilde{D}_{\alpha\beta}^2, \bigvee_{\alpha\beta} \partial \tilde{D}_{\alpha\beta}^2) \quad \text{since } \tilde{X} \text{ is contractible} \\ &\approx \pi_1(\bigvee_{\alpha\beta} \partial \tilde{D}_{\alpha\beta}^2) \quad \text{since } \bigvee_{\alpha\beta} \tilde{D}_{\alpha\beta}^2 \text{ is contractible} \end{aligned}$$

This last group is free with basis the loops $\partial \tilde{D}_{\alpha\beta}^2$, so the inclusions $\tilde{D}_{\alpha\beta}^2 \hookrightarrow \bigvee_{\alpha\beta} \tilde{D}_{\alpha\beta}^2$ form a basis for $\pi_2(\bigvee_{\alpha\beta} \tilde{D}_{\alpha\beta}^2, \bigvee_{\alpha\beta} \partial \tilde{D}_{\alpha\beta}^2)$. This implies that $\pi_2(Y, Y^1)$ is generated by the inclusions $D_\alpha^2 \hookrightarrow Y$ and their images under the action of loops in X . The same is true for $\pi_2(W, Z)$ via the surjection $\pi_2(Y, Y^1) \rightarrow \pi_2(W, Z)$. Using the isomorphism $\pi_2(W, Z) \approx \pi_2(W, X)$, we conclude that $\pi_2(W, X)$ is generated by the characteristic maps of the cells e_α^2 and their images under the action of $\pi_1(X)$. $\square$

Proof of the general Hurewicz Theorem: As in the earlier form of the theorem we may assume (X, A) is a CW pair such that the cells of $X - A$ have dimension $\geq n$.

We first prove the theorem assuming that $\pi_1(A) \rightarrow \pi_1(X)$ is an isomorphism. This is always the case if $n \geq 3$, so this case will finish the proof except when $n = 2$. We may assume also that $X = X^{n+1}$ since higher-dimensional cells have no effect on π_n or H_n . Consider the commutative diagram

$$\begin{array}{ccccccc} \pi_{n+1}(X, X^n \cup A) & \xrightarrow{\partial} & \pi_n(X^n \cup A, A) & \xrightarrow{i_*} & \pi_n(X, A) & \longrightarrow & 0 \\ \downarrow & & \downarrow & & \downarrow & & \\ \pi'_{n+1}(X, X^n \cup A) & \xrightarrow{\partial'} & \pi'_n(X^n \cup A, A) & \xrightarrow{i'_*} & \pi'_n(X, A) & \longrightarrow & 0 \\ \downarrow h' & & \downarrow h' & & \downarrow h' & & \\ H_{n+1}(X, X^n \cup A) & \longrightarrow & H_n(X^n \cup A, A) & \longrightarrow & H_n(X, A) & \longrightarrow & 0 \end{array}$$

The first and third rows are exact sequences for the triple $(X, X^n \cup A, A)$. The left-hand h' is an isomorphism since by the preceding lemma, $\pi_{n+1}(X, X^n \cup A)$ is a free π_1 -module with basis the characteristic maps of the $(n+1)$ -cells of $X - A$, so $\pi'_{n+1}(X, X^n \cup A)$ is a free abelian group with the same basis, and $H_{n+1}(X, X^n \cup A)$ is also free with basis the $(n+1)$ -cells of $X - A$. Similarly, the lemma implies that

the middle h' is an isomorphism since the assumption that $\pi_1(A) \rightarrow \pi_1(X)$ is an isomorphism implies that $\pi_1(A) \rightarrow \pi_1(X^n \cup A)$ is injective, hence an isomorphism if $n \geq 2$.

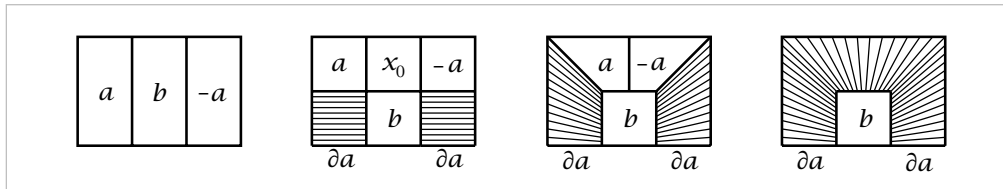
A simple diagram chase now shows that the right-hand h' is an isomorphism. Namely, surjectivity follows since $H_n(X^n \cup A, A) \rightarrow H_n(X, A)$ is surjective and the middle h' is an isomorphism. For injectivity, take an element $x \in \pi'_n(X, A)$ with $h'(x) = 0$. The map i'_* is surjective since i_* is, so $x = i'_*(y)$ for some element $y \in \pi'_n(X^n \cup A, A)$. Since the first two maps h' are isomorphisms and the bottom row is exact, there is a $z \in \pi'_{n+1}(X, X^n \cup A)$ with $\partial'(z) = y$. Hence $x = 0$ since $i_*\partial = 0$ implies $i'_*\partial' = 0$.

It remains to prove the theorem when $n = 2$ and $\pi_1(A) \rightarrow \pi_1(X)$ is not an isomorphism. The proof above will apply once we show that the middle h' in the diagram is an isomorphism. The preceding lemma implies that $\pi'_2(X^2 \cup A, A)$ is generated by characteristic maps of the 2-cells of $X - A$. The images of these generators under h' form a basis for $H_2(X^2 \cup A, A)$. Thus h' is a homomorphism from a group which, by the lemma below, is abelian to a free abelian group taking a set of generators to a basis, hence h' is an isomorphism. $\square$

Lemma 4.39. *For any (X, A, x_0) , the formula $a + b - a = (\partial a)b$ holds for all $a, b \in \pi_2(X, A, x_0)$, where $\partial: \pi_2(X, A, x_0) \rightarrow \pi_1(A, x_0)$ is the usual boundary map and $(\partial a)b$ denotes the action of ∂a on b . Hence $\pi'_2(X, A, x_0)$ is abelian.*

Here the '+' and '-' in $a + b - a$ refer to the group operation in the nonabelian group $\pi_2(X, A, x_0)$.

Proof: The formula is obtained by constructing a homotopy from $a + b - a$ to $(\partial a)b$ as indicated in the pictures below. $\square$



The Plus Construction

There are quite a few situations in algebraic topology where having a nontrivial fundamental group complicates matters considerably. We shall next describe a construction which in certain circumstances allows one to modify a space so as to eliminate its fundamental group, or at least simplify it, without affecting homology or cohomology. Here is the simplest case: